

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S9 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($RFS \leq DMFS$), and a patient cannot die without having both events ($DMFS \leq OS$). The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic



Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

2.2 Observed vs. Fitted Survival Curves

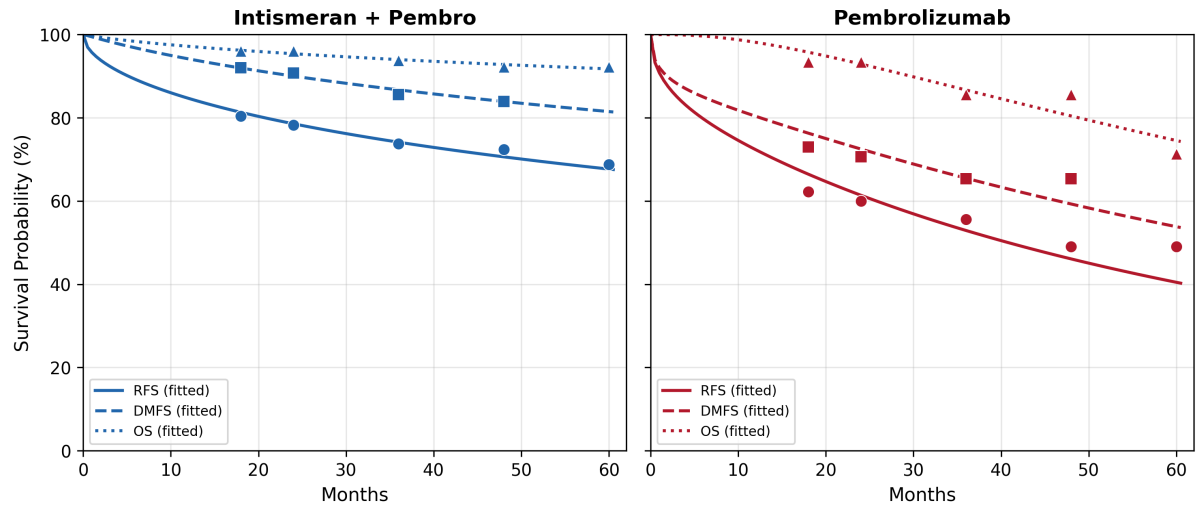


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

3 Survival Model Selection

3.1 Component-Level Parametric Fit

Table S3: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

3.2 Joint Model Selection

A total of 4,096 distribution combinations were evaluated for structural validity at 40 years. All 4,096 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_4096.csv](#)).

Table S4: Top 10 Joint Model Combinations by Total AIC

Rank	OS	r_{DM}	r_{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / LL	LL / W	-222.61	76.3%	0.0%
2	LN / W	LN / LL	LL / LL	-222.52	76.3%	0.0%
3	LN / W	Exp / LL	LL / W	-222.47	76.3%	0.0%
4	LLog / W	LN / LL	LL / W	-222.42	72.9%	0.0%
5	LN / W	LN / LL	LN / W	-222.40	76.3%	0.0%
6	LN / W	LN / LL	LL / LN	-222.39	76.3%	0.0%
7	LN / W	Exp / LL	LL / LL	-222.38	76.3%	0.0%
8	W / W	LN / LL	LL / W	-222.37	71.1%	0.0%
9	LLog / W	LN / LL	LL / LL	-222.33	72.9%	0.0%
10	LN / W	LN / LL	LN / LL	-222.31	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3%, which is substantially higher than GP survival at age 101 ($\approx 0\%$). This motivated the GP mortality constraint.

3.3 Lifetime OS Extrapolation by Distribution

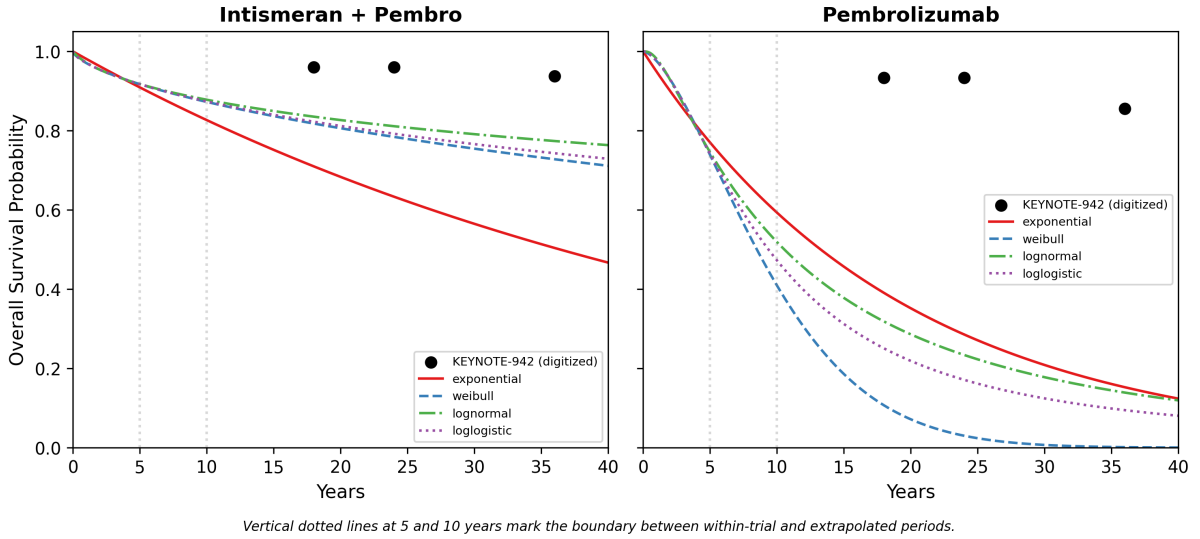


Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four distributions produce wildly divergent 40-year survival estimates (0% to 76.3%).

4 General-Population Mortality and Structural Validation

4.1 GP Mortality Implementation

The GP mortality constraint caps modeled OS at the age- and sex-matched US general population survival from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:**

Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$. **Constraint formula (mortality hazard floor):** from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{GP}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

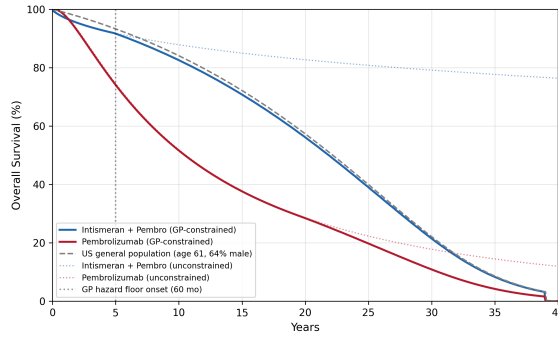
where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{GP}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

4.2 Long-Term Survival Diagnostics

Table S5: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival	Combo vs GP
5	91.6%	74.1%	93.3%	Below (excess mortality to date)
10	82.5%	51.6%	84.1%	Below (hazard floor active)
15	70.8%	37.6%	72.2%	Below (hazard floor active)
20	56.1%	28.4%	57.3%	Below (hazard floor active)
30	21.5%	10.9%	22.1%	Below (hazard floor active)
40	0.0%	0.0%	0.0%	At GP (age 101, GP survival ≈ 0)

4.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

4.4 Structural Quality Assurance

Table S6: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

5 External Validation

5.1 External Validation Landmarks

Table S7: External Validation of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	49.1%	~50% (nivo)
RFS	7	31.9%	50.0% (46–55%)	~44% (nivo)
DMFS	5	53.8%	~58%	58% (nivo)
DMFS	7	44.9%	54% (50–59%)	~50% (nivo)
OS	5	74.5%	NR	76% (nivo)
OS	7	64.2%	NR	~70% (nivo)
RFS	10	23.2%	~47% (extrap.)	~40% (nivo, est.)
OS	10	51.9%	NR	~65% (nivo, est.)

NR: Not reported. nivo: CheckMate 238 evaluated nivolumab. The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population. OS is consistent with CheckMate 238.

5.2 Longitudinal External Validation

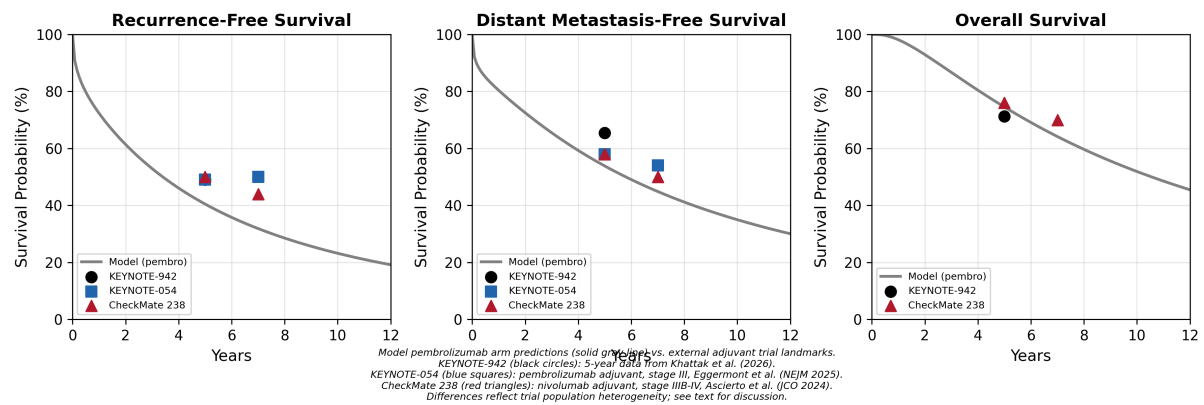


Figure S5: External validation of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942. Squares: KEYNOTE-054. Triangles: CheckMate 238.

6 Model Inputs

6.1 Complete Parameter Table

Table S8: Full Model Parameter Inventory

Parameter	Base	DSA Low	DSA High	PSA Distribution
<i>Clinical Parameters</i>				
OS μ – combo	4.428	4.228	4.628	Normal (SE from Jac)
OS σ – combo	0.478	0.378	0.578	Normal (SE from Jac)
OS μ – pembro	3.725	3.525	3.925	Normal (SE from Jac)
OS σ – pembro	0.574	0.474	0.674	Normal (SE from Jac)
r _{DM} μ – combo	4.107	3.907	4.307	Normal (SE from Jac)
r _{DM} σ – combo	0.432	0.332	0.532	Normal (SE from Jac)
r _{DM} μ – pembro	2.849	2.649	3.049	Normal (SE from Jac)
r _{DM} σ – pembro	0.934	0.834	1.034	Normal (SE from Jac)
r _{LR} μ – combo	3.367	3.167	3.567	Normal (SE from Jac)
r _{LR} σ – combo	0.624	0.524	0.724	Normal (SE from Jac)
r _{LR} μ – pembro	2.750	2.550	2.950	Normal (SE from Jac)
r _{LR} σ – pembro	0.638	0.538	0.738	Normal (SE from Jac)
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed
Baseline age	61 years	55	67	Fixed
Male proportion	64%	–	–	Fixed
<i>Costs (2026 USD)</i>				
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Gamma (CV=0.2)
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.2)
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.2)
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.2)
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)
<i>Utilities</i>				
RF state	0.83	0.80	0.86	Beta (SE=0.03)
LR state	0.64	0.61	0.67	Beta (SE=0.03)
DM state	0.55	0.52	0.58	Beta (SE=0.03)
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)
<i>Economic Parameters</i>				
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)
Time horizon	40 years	10	20	Fixed (scenario)
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)
Cost base year	2026 USD	–	–	–
GP mortality source	US 2019 Life Tables	–	–	–

7 Additional Results

7.1 Lifetime Cost Decomposition

Table S9: Lifetime Cost Decomposition by Component

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$493	\$493	\$0
Adverse events	\$15,000	\$0	\$15,000
LR management	\$83,196	\$69,500	\$13,696
DM management	\$342,542	\$425,276	-\$82,734
Total	\$860,118	\$713,157	\$146,961

The upfront \$200,000 intismeran cost is partially offset by \$82,734 in lower DM management costs, reflecting the DMFS HR of 0.411.

7.2 LY and QALY Decomposition

Table S10: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	20.99	13.86	7.13
Life-years (discounted 3%)	14.71	10.29	4.42
RF state QALYs	8.32	4.49	3.83
LR state QALYs	1.48	1.24	0.24
DM state QALYs	1.31	1.62	-0.32
Total QALYs	11.10	7.34	3.76

Health benefit is driven by RF state QALYs (3.83 gained), more than compensating for fewer DM-state QALYs (-0.32).

8 Uncertainty and Structural Scenario Analyses

8.1 Deterministic Sensitivity Analysis

Table S11: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	40,601–37,715	\$27,691
Utility LR	0.64	0.61	0.67	39,224–38,987	\$2,283
Utility DM	0.55	0.52	0.58	38,926–39,285	\$3,447
Pembrolizumab annual cost	\$220,896	176,717	265,075	39,105–39,105	\$0
Intismeran cost	\$200,000	160,000	240,000	28,461–49,749	\$80,000
LR monthly cost	\$3,000	2,400	3,600	38,376–39,834	\$5,478
DM monthly cost	\$12,000	9,600	14,400	43,508–34,702	\$33,094
Discount rate	3%	0%	5%	31,092–47,814	\$257,766
OS μ combo	4.428	4.228	4.628	DQY < 0 at low	\$37,797
OS μ pembro	3.725	3.525	3.925	59,135–57,130	\$30,496

The ICER remained below \$60,000/QALY across all tested ranges.

8.2 Probabilistic Sensitivity Analysis

8.2.1 PSA Implementation Details

PSA with 2,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$). Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints (RFS $\leq$ DMFS $\leq$ OS, monotonicity, survival within [0,1]) verified every cycle; all 2,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Table S12: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sample
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Jo
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Jo
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Jo
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000	2.278 / 3.130	0.995	Jo
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000	2.039 / 1.928	0.997	Jo
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Jo
Intismeran cost	Gamma (CV=0.2)	\$200,000	—	—	Indepe
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Indepe
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Indepe
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Indepe
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Indepe

8.2.2 PSA Results

Mean Δ QALY = 4.03 (95% CI 2.33–6.74), mean Δ Cost = \$121,796 (95% CI \$-610,346 to \$509,272). Mean NMB = \$281,051 at \$100K, \$482,475 at \$150K. P(CE) = 63.9% at \$50K, 94.9% at \$100K, 98.7% at \$150K. No iterations were excluded based on the ICER value; 17.6% of draws were cost-saving.

8.2.3 PSA Convergence

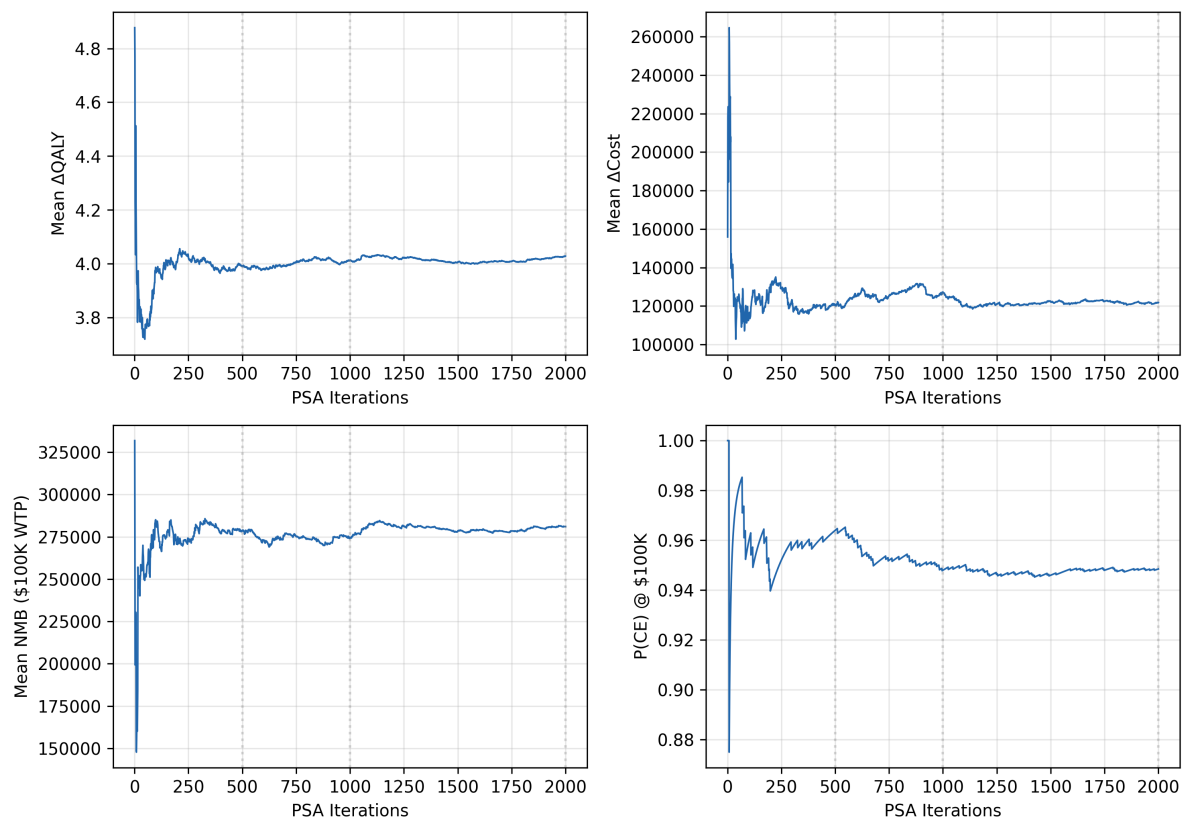


Figure S6: PSA convergence across iteration counts (2,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of $\Delta QALY$, $\Delta Cost$, NMB, and P(CE). Vertical dotted lines at 500, 1,000, and 2,000 iterations. Estimates stable from ~ 500 iterations.

8.2.4 Cost-Effectiveness Plane

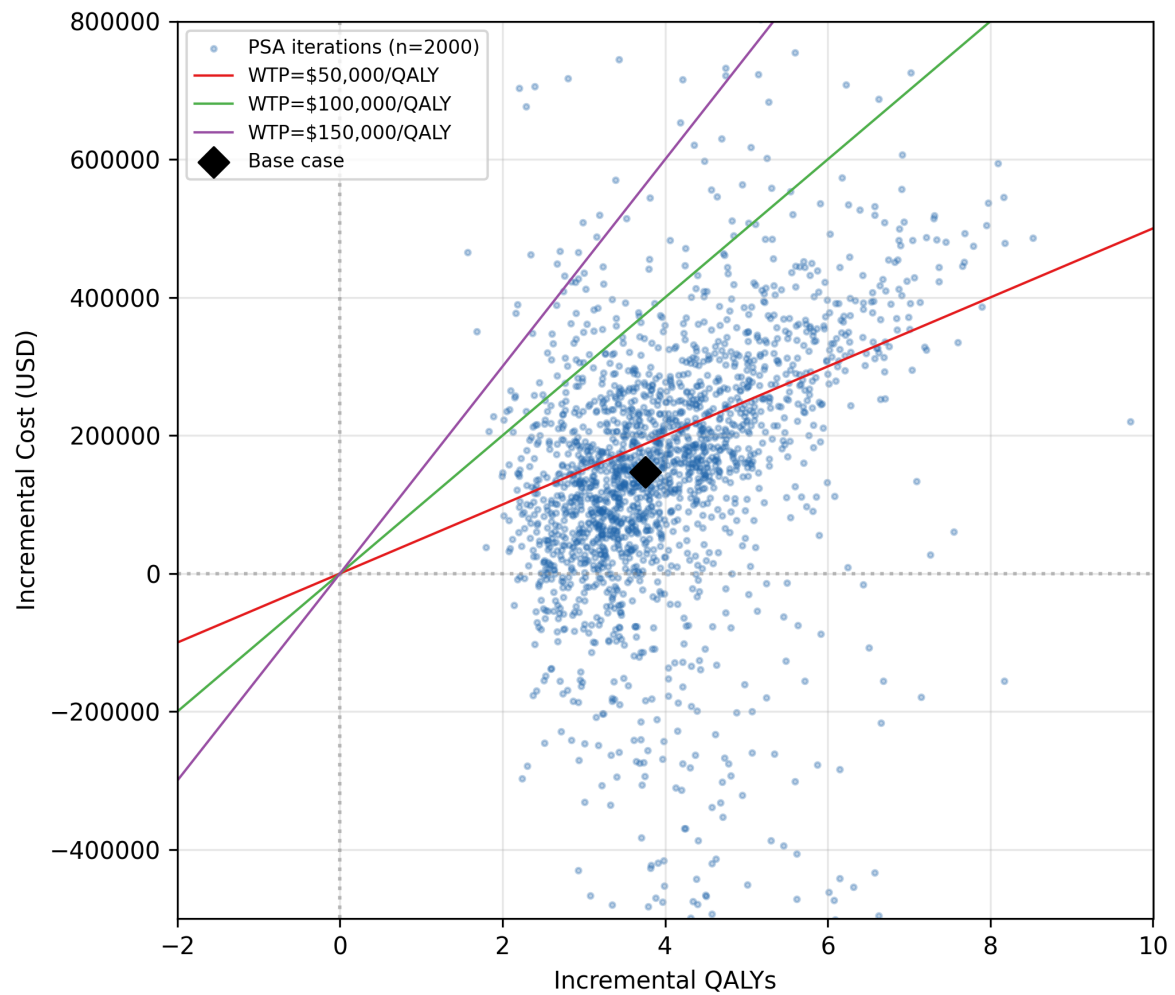


Figure S7: CE plane showing 2,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 17.6% of iterations fall in the southeast quadrant (cost-saving). No iterations were excluded based on the ICER value.

8.3 Structural Scenario Analyses

Table S13: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Weibull OS	6.77	5.35	\$293,679	\$54,921	\$241,047	\$508,4
Unconstrained (no GP)	9.24	7.22	\$373,488	\$51,753	\$348,189	\$709,0
GP floor 48 mo	4.37	3.72	\$145,054	\$39,014	\$226,750	\$412,6
GP floor 60 mo (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
GP floor 72 mo	4.50	3.81	\$149,673	\$39,251	\$231,648	\$422,3
10-year horizon	1.27	1.28	\$95,573	\$74,945	\$31,951	\$95,7
20-year horizon	3.32	2.91	\$115,739	\$39,734	\$175,548	\$321,1
40-year horizon (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Intismeran \$100K	4.42	3.76	\$46,961	\$12,496	\$328,851	\$516,7
Intismeran \$200K (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Intismeran \$300K	4.42	3.76	\$246,961	\$65,714	\$128,851	\$316,7
Intismeran \$500K	4.42	3.76	\$446,961	\$118,932	-\$71,149	\$116,7
Waning V1 (recurrence)	1.83	1.87	\$38,544	\$20,591	\$148,647	\$242,2
Waning V2 (hazard convergence)	2.66	2.47	\$74,620	\$30,176	\$172,659	\$296,2
No direct OS benefit	0.00	0.50	-\$14,562	Dominant	\$64,131	\$88,9

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

9 Value-Based Pricing

9.1 Price Threshold Analysis

Table S14: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	-\$14,113	\$428,851	\$616,757
\$50,000	-\$809	\$378,851	\$566,757
\$100,000	\$12,496	\$328,851	\$516,757
\$150,000	\$25,800	\$278,851	\$466,757
\$200,000 (base)	\$39,105	\$228,851	\$416,757
\$250,000	\$52,409	\$178,851	\$366,757
\$300,000	\$65,714	\$128,851	\$316,757
\$350,000	\$79,019	\$78,851	\$266,757
\$400,000	\$92,323	\$28,851	\$216,757
\$429,077	\$100,060	-\$226	\$187,680
\$450,000	\$105,628	-\$21,149	\$166,757
\$500,000	\$118,932	-\$71,149	\$116,757
\$550,000	\$132,237	-\$121,149	\$66,757
\$600,000	\$145,541	-\$171,149	\$16,757
\$616,455	\$149,920	-\$187,604	\$302
\$650,000	\$158,846	-\$221,149	-\$33,243
\$700,000	\$172,150	-\$271,149	-\$83,243
\$750,000	\$185,455	-\$321,149	-\$133,243
\$800,000	\$198,759	-\$371,149	-\$183,243

Max price @ \$100K/QALY: **\$429,077** ($2.1 \times$ base). Max price @ \$150K/QALY: **\$616,455** ($3.1 \times$ base).

9.2 ICER versus Acquisition Price

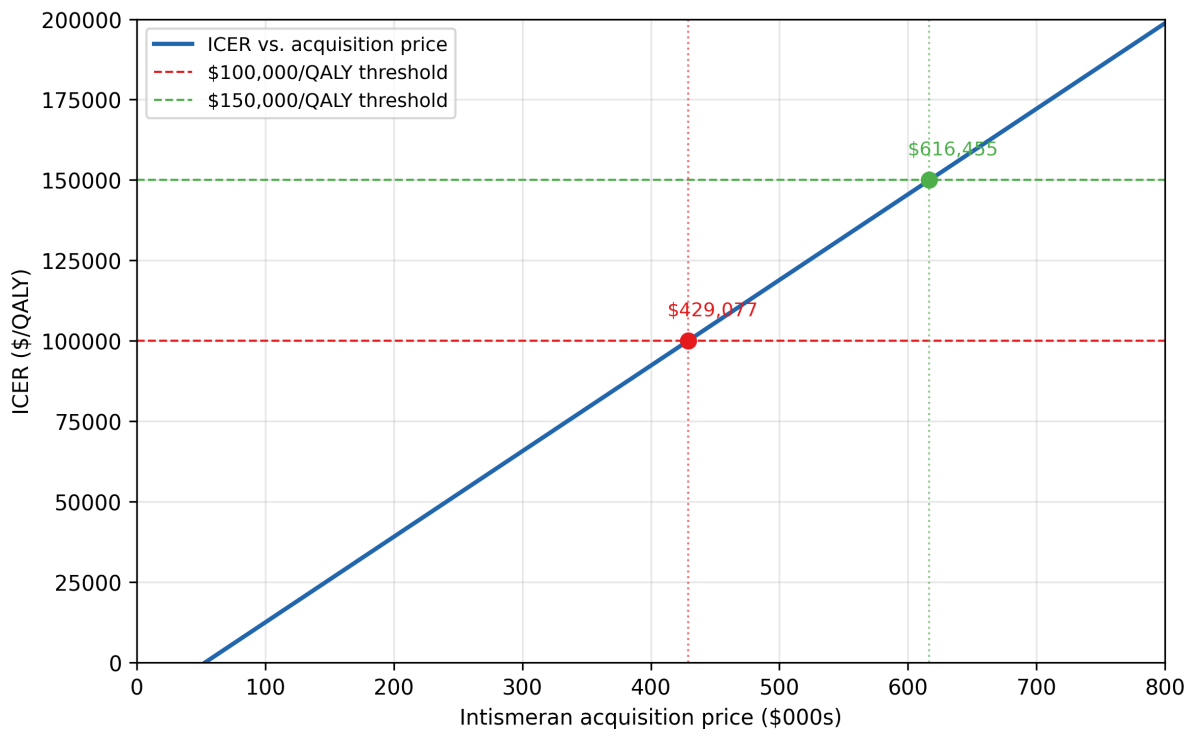


Figure S8: ICER as a function of intismeran acquisition price. Red/green dashed lines: \$100K/\$150K thresholds. Vertical dotted lines: threshold prices.

10 Reproducibility and Reporting

10.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: continuous-time exponential: $1/(1+r)^t$ where $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling

- `survival_model_selection.py` — 4,096-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing
- `figures.py` / `gen_fig*.py` — Figure generation scripts
- `output/survival_combos_4096.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa_full.py` reproduces PSA. GitHub tag `v1.0-submission` freezes the submission code.

10.2 CHEERS 2022 Checklist

Table S15: CHEERS 2022 Checklist

No.	Item	Reported Location	
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	Yes	Methods
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods
19	Analytic methods	Yes	Methods
20	Study parameters	Yes	Methods, Appendix D
21	Incremental results	Yes	Results
22	Uncertainty	Yes	PSA, DSA
23	Heterogeneity	Yes	Subgroup
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

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